

Week 2 Notes

STAT 251 SECTION 01

Describing and Visualizing Distributions

As we learned in the previous lecture, descriptive statistics is the first step to statistical analysis and involves the summary and organization of data. One of the main ways we can summarize data is to describe the characteristics of the **distribution** of the variable(s) in the data. A distribution tells us something about what kinds of values a variable can have and how often they occur.

Data used this week

Each dataset below is a plain `.csv`, ready to read straight into R or to upload to the [tools and calculators \(https://stat-tools.jmks-homelab.dev/\)](https://stat-tools.jmks-homelab.dev/).

- [Teen cell-phone survey \(./Data/teens_cell_phone_focus_simulated.csv\)](#) — 743 responses (Figure 1 below).
- [Old Faithful eruptions \(./Data/old_faithful.csv\)](#) — 272 eruptions, duration and waiting time.
- [MPG and engine cylinders \(./Data/carsdata.csv\)](#) — 32 car models, used for the dot plot examples in class.
- Consider the following example: Are teens distracted by their cell phones? A study conducted by the Pew Research center surveyed 743 U.S teenagers ages 13-17 to understand how cell phones in class impacted their ability to concentrate. Each student was asked to rate the impact of cell phone use on their concentration based on a Likert scale. The data and a plot of the distribution of survey responses are given below

Survey data layout: the first four teens and the last, of 743.

Student	Age	Response
1	13	Never
2	13	Sometimes
3	15	Never

Student	Age	Response
4	17	Often
⋮	⋮	⋮
743	16	Rarely

- What type of variable is *Age*?
- What type of variable is *Response*?

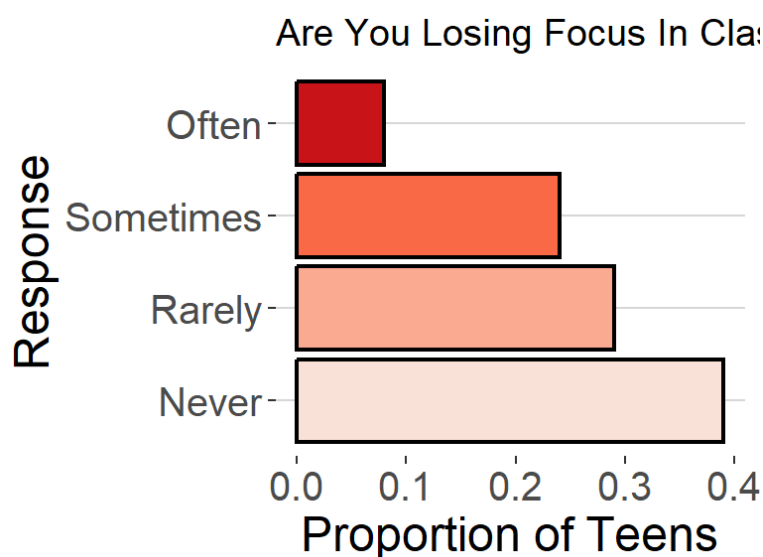


Figure 1. Distribution of student responses to the survey question “Are you losing focus in class by checking your cell phone?”

Download the data behind Figure 1 ([./Data/teens_cell_phone_focus_simulated.csv](#)) — one row per student. Pew published only the four percentages, so the individual rows are simulated; the response counts (289 Never, 216 Rarely, 178 Sometimes, 60 Often, $n = 743$) are the published ones and reproduce Figure 1 exactly.

- Looking at the distribution of survey responses above, would you say that most teens feel they are distracted by their cell phones?

Distributions

Formally, a **distribution** is a function that gives (a) the possible values of a variable and (b) how often each value occurs. The *how often* is usually described using *frequency* or *relative frequency*.

- The **frequency** of a given value is simply a count of the number of times that the given values occurs
- The **relative frequency** of a given value is the proportion of times that the given value occurs
- We can also use the *cumulative relative frequency* to describe how often a variable occurs. The **cumulative relative frequency** is the proportion of values that are *less than or equal to* a given value in a distribution.
 - **Note:** As we said with the example about distracted teens, we can talk about the distribution of a categorical variable as well, however we *may not use* the cumulative relative frequency if there is no natural order to the values (i.e qualitative - nominal).

Consider the following hypothetical set of 10 exam scores (out of 10 points) for students in a statistics course

$$\{5, 6, 6, 7, 7, 7, 8, 9, 9, 10\}$$

- the possible test scores are 5, 6, 7, 8, 9, and 10. Lets try computing the frequency, relative frequency, and cumulative relative frequency for students who score a 6 on the exam.
 - The frequency for an exam score of 6 is $freq(6) = 2$ because 2 of the 10 students had an exam score of 6.
 - Similarly, the relative frequency for an exam score of 6 is $2/10 = 0.2$.
 - The cumulative relative frequency for an exam score of 6 will be the total number of values in the data that are less than or equal to the value 6. This will be the frequency of 5's plus the frequency of 6's which will be $(1 + 2)/10 = 3/10$ or 0.3. Alternatively we could simply add the relative frequency for an exam score of 5 to the relative frequency for an exam score of 6: $0.1 + 0.2 = 0.3$

Frequency table for the ten exam scores.

Score	Frequency	Relative Frequency	Cumulative Relative Frequency
5	1	0.1	0.1
6	2	0.2	0.3
7	3	0.3	0.6
8	1	0.1	0.7
9	2	0.2	0.9

Score	Frequency	Relative Frequency	Cumulative Relative Frequency
10	1	0.1	1.0

- The table above is called a **frequency table**. It is a tabular display of the distribution of a variable.

A frequency table is only one way of describing a distribution, and it is the plainest. The rest of these notes introduce the other ways: how to build a frequency table when the variable is continuous, what features of a distribution we look for, and which graph suits which kind of variable.

Frequency tables for continuous variables

Every frequency table above listed the possible values of the variable in its first column and counted them in the rest. That works whenever the values can be listed.

We can create a frequency table for any type of variable. However, when dealing with quantitative continuous variables we cannot list all possible values because the number of possible values is infinite. Therefore, we usually have to make a decision about how we want to group values of the variable, a process called binning. **Binning** is a way to group numbers of more-or-less continuous values into a smaller number of "bins". These bins represent a collection of values that fall into a certain range and make listing the possible values a lot easier.

Consider the following trivial example using the quantitative continuous variable

$$X = \{1.23, 1.38, 1.76, 2.60, 2.65, 2.79, 2.89, 3.12, 3.15, 3.18, 3.65, 3.77, 3.87, 4.12, 4.41\}$$

Each observation of the variable X is measured with a precision of two decimal places, and no two observations in the sample are identical. As a result if we simply listed the possible values we would have 15 unique values each with a frequency of 1 which would produce a frequency table that wasn't very informative. We can, instead, use binning to extract more meaningful information about the distribution of X . Now consider the frequency table using binning:

Frequency table for X using seven bins of width 0.5.

Bin	Frequency	Relative Frequency	Cumulative Relative Frequency
$1 < X \leq 1.5$	2	0.13	0.13
$1.5 < X \leq 2$	1	0.07	0.20
$2 < X \leq 2.5$	0	0.00	0.20

Bin	Frequency	Relative Frequency	Cumulative Relative Frequency
$2.5 < X \leq 3$	4	0.27	0.47
$3 < X \leq 3.5$	3	0.20	0.67
$3.5 < X \leq 4$	3	0.20	0.87
$4 < X \leq 4.5$	2	0.13	1.00

The following is another example using $n = 272$ observations collected on the waiting time between eruptions and the duration of the eruption of the Old Faithful geyser in Yellowstone National Park, Wyoming, USA.

[Download the Old Faithful data \(./Data/old_faithful.csv\)](#) (all 272 eruptions).

The first eight Old Faithful eruptions,
of 272.

Duration (min)	Waiting Time (min)
3.6	79
1.8	54
3.333	74
2.283	62
4.533	85
2.883	55
4.7	88
3.6	85
$\vdots$	$\vdots$

Consider a frequency table to describe the distribution of the variable *waiting time (min)*:

Frequency table for Old Faithful waiting time using twelve bins of width 5 minutes.

Waiting Time	Frequency	Relative Frequency	Cumulative Relative Frequency
$40 < X \leq 45$	4	0.015	0.015

Waiting Time	Frequency	Relative Frequency	Cumulative Relative Frequency
$45 < X \leq 50$	22	0.081	0.096
$50 < X \leq 55$	33	0.121	0.217
$55 < X \leq 60$	24	0.088	0.305
$60 < X \leq 65$	14	0.051	0.357
$65 < X \leq 70$	10	0.037	0.393
$70 < X \leq 75$	27	0.099	0.493
$75 < X \leq 80$	54	0.199	0.691
$80 < X \leq 85$	55	0.202	0.893
$85 < X \leq 90$	23	0.085	0.978
$90 < X \leq 95$	5	0.018	0.996
$95 < X \leq 100$	1	0.004	1.000

- How long does a visitor typically have to wait before seeing old faithful erupt?
- Is there more than one “typical” value in this distribution?

IMPORTANT: one thing to consider is that grouping continuous data into discrete intervals leads to a loss of precision. Sometimes this can result in the loss of valuable information present in the original data. In addition, how we choose to bin the data (i.e how wide we make the intervals) may lead to different, sometimes erroneous, conclusions about the distribution and patterns within the data.

Consider the frequency table for the *waiting time (min)* with only two bins

The same waiting times collapsed into only two bins of width 30 minutes.

Waiting Time	Frequency	Relative Frequency	Cumulative Relative Frequency
$40 < X \leq 70$	107	0.393	0.393
$70 < X \leq 100$	165	0.607	1.000

- How does using only two bins/intervals alter your conclusion regarding the typical time a visitor must wait before witnessing Old Faithful’s eruption?

Features of a distribution

Descriptive statistics or graphical summaries can tell us a lot about the variables in our data. Graphs help us visualize trends in the data, identify problematic observations, and can help inform the types of analyses we should use. We may be interested in certain features related to the distribution(s) of the variable(s) in our data, such as the *shape*, *center*, and *variability*.

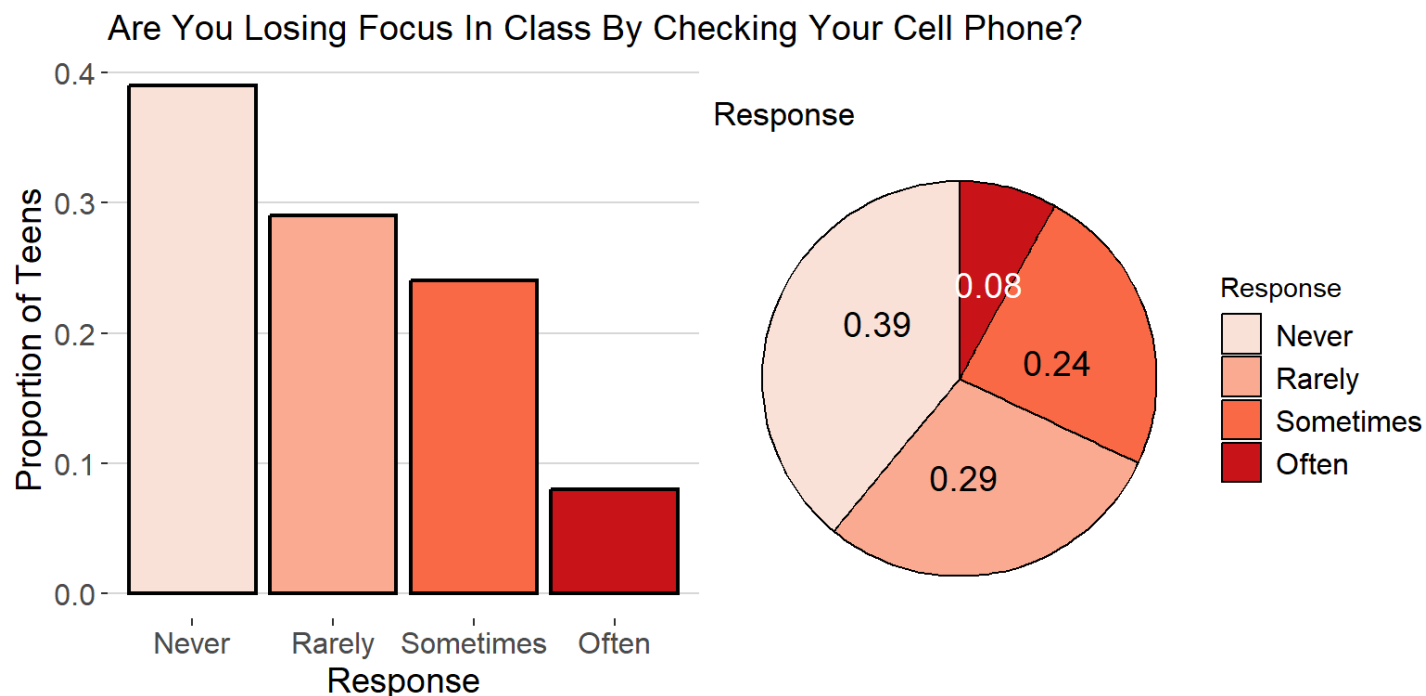
- **shape** refers to how observations cluster across the range of a variable. Certain aspects to note would be where observations are densely packed (values with high frequencies) and where observations are sparser (values with low frequencies). This can give us a feel for where a typical value would appear and what values may be considered “rare” or “extreme”.
- **Center** refers to the “middle point” on the distribution and also supplies information about where a typical value falls.
- **Variability** refers to how tightly observations cluster around the center of a distribution. When values are tightly packed they are said to have *low variability* vs when they are spread out they are said to have *high variability*
- **modal category** is a feature of qualitative data only and refers to the most frequent category of a qualitative variable.

Graphs for qualitative data

There are several graphs that can be used to display qualitative data. The most common are bar graphs and pie charts.

- A **Bar graph** is simple and widely used plot for displaying the distribution of categorical variables. Each category is represented by a bar, and the height of the bar corresponds to the frequency or proportion of observations in that category. When the bars are sorted in decreasing frequency it is called a **Pareto chart**.
- A **Pie chart** is useful for illustrating the proportional contribution of each category to the whole. Beyond this, it is not a particularly useful plot when describing the features of a distribution.
- Ex. Recall the Pew Research survey above, asking teens to rate how distracted they are by their cell phones. The following are a bar graph and pie chart of student responses

Download the response data ([./Data/teens_cell_phone_focus_simulated.csv](#)) (simulated student-level rows reproducing the published Pew counts).



- what is the modal category of the variable *Response*?

Graphs for quantitative data

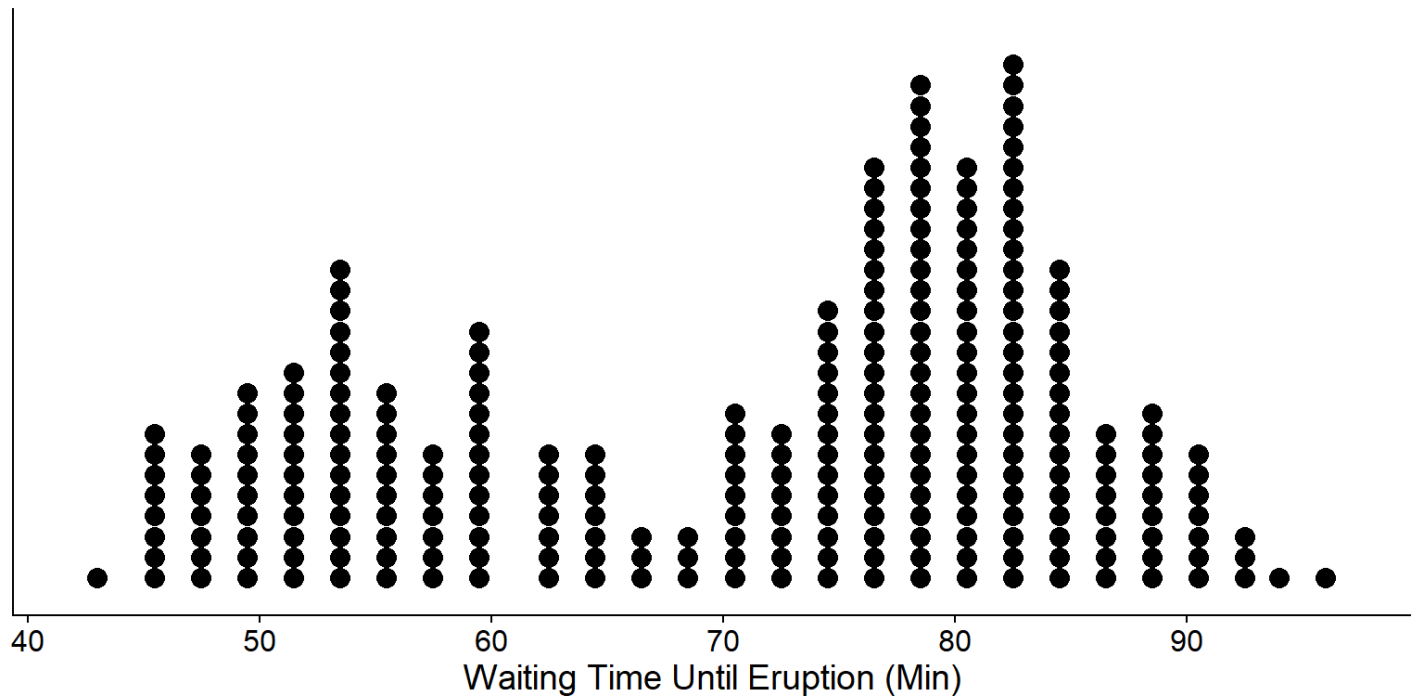
There are also many plots that can be used with quantitative data. Some of the most commonly used graphs are listed below.

In class we build these plots from the [MPG and engine-cylinder data for 32 car models](#) ([./Data/carsdata.csv](#)).

- A **dot plot** shows a dot for each observation placed just above the value on the number line for that observation. To construct a dot plot:
 - 1. Draw a horizontal line, label it with the name of variable, and mark regular values of the variable on the line.
 - 2. For each observation, place a dot above its value on the number line.
- A **stem and leaf plot** is similar to a dot plot in that it displays individual observations of a variable. Each observation in the plot consists of a stem and a leaf. The *stem* is all of the digits except for the last digit. The *leaf* is the last digit. For example, take the observation 12.3 the stem for this observation is 12 and the leaf is 3. To construct a stem and leaf plot:
 - 1. Draw a table with two columns and label the left column "Stem" and the right column "leaf".

- 2. Next, sort the observations from smallest to largest and place the stems in the left column of your table in increasing order.
- 3. in the right column, fill in the leaves of each stem in increasing order.

Example: A dot plot constructed for the $n = 272$ observations collected on the waiting time between eruptions and the duration of the eruption of the Old Faithful geyser in Yellowstone National Park, Wyoming, USA.



- Think about the shape of this distribution and how it relates to the time that a typical visitor can expect to wait before the geyser erupts. Is the result consistent with the frequency table we constructed earlier?

Example: A stem and leaf plot constructed for the $n = 272$ observations collected on the waiting time between eruptions and the duration of the eruption of the Old Faithful geyser in Yellowstone National Park, Wyoming, USA.

Read the plot below one row at a time. The number to the left of the bar is the stem, here the tens digit of the waiting time; each digit to the right of the bar is one leaf, the units digit of a single eruption. A row reading `5 | 0011` would stand for four eruptions, at 50, 50, 51 and 51 minutes.

Note that R has *split* each stem across two rows, so every tens digit appears twice: the first row collects the leaves 0 through 4, the second the leaves 5 through 9. The two rows beginning `5` therefore cover 50–54 minutes and 55–59 minutes respectively.

The longer a row is, the more eruptions fell in that five-minute span. That is what makes the shape audible as well as visible: the rows grow long in the low fifties, shrink to their shortest in the middle sixties, then grow longest of all in the high seventies and low eighties — the same two clusters, and

the same gap between them, that the dot plot showed.

```
##
##   The decimal point is 1 digit(s) to the right of the |
##
##   4 | 3
##   4 | 55566666777788899999
##   5 | 00000111111222223333333444444444
##   5 | 555555666677788889999999
##   6 | 00000022223334444
##   6 | 555667899
##   7 | 0000111112333333344444
##   7 | 55555556666666667777777777788888888888889999999999
##   8 | 0000000011111111111122222222223333333333334444444444
##   8 | 55555566666677888888999
##   9 | 00000012334
##   9 | 6
```

Both dot plots and stem and leaf plots are excellent choices for discrete variables with a small to moderate number of observations. However, when a variable consists of many observations, has low variability, or is continuous, it can be difficult to construct these two plots.

- When dealing with continuous data, it is often necessary to round observations to the nearest tenth before constructing either plot.

The following are alternatives that work for variables of any number of observations.

More graphs for quantitative data

- A **Histogram** is a specialized type of bar graph. It uses bars to portray the frequencies or relative frequencies of the possible outcomes for a quantitative variable of either type. The histogram uses binning to condense the view of a quantitative variable into a smaller set of intervals which makes it particularly useful when the number of observations is large. The width of the bins is arbitrary but all bins should be the same length and cover all possible values of the data. The steps for constructing a histogram are:
 - 1. Divide the data into non-overlapping intervals (bins) of equal length.
 - 2. Compute the frequency of each interval (this is the same as constructing the frequency table for the chosen bins)

- 3. Label the x-axis with the midpoints or endpoints of each interval.
- 4. Draw a bar over each value or interval with height equal to its frequency or relative frequency
- In general, you can choose the number of bins k you wish to use and set the bin width w accordingly or alternatively set the bin width you want and use the corresponding number of bins.

- formula for choosing the number of bins based on bin width w

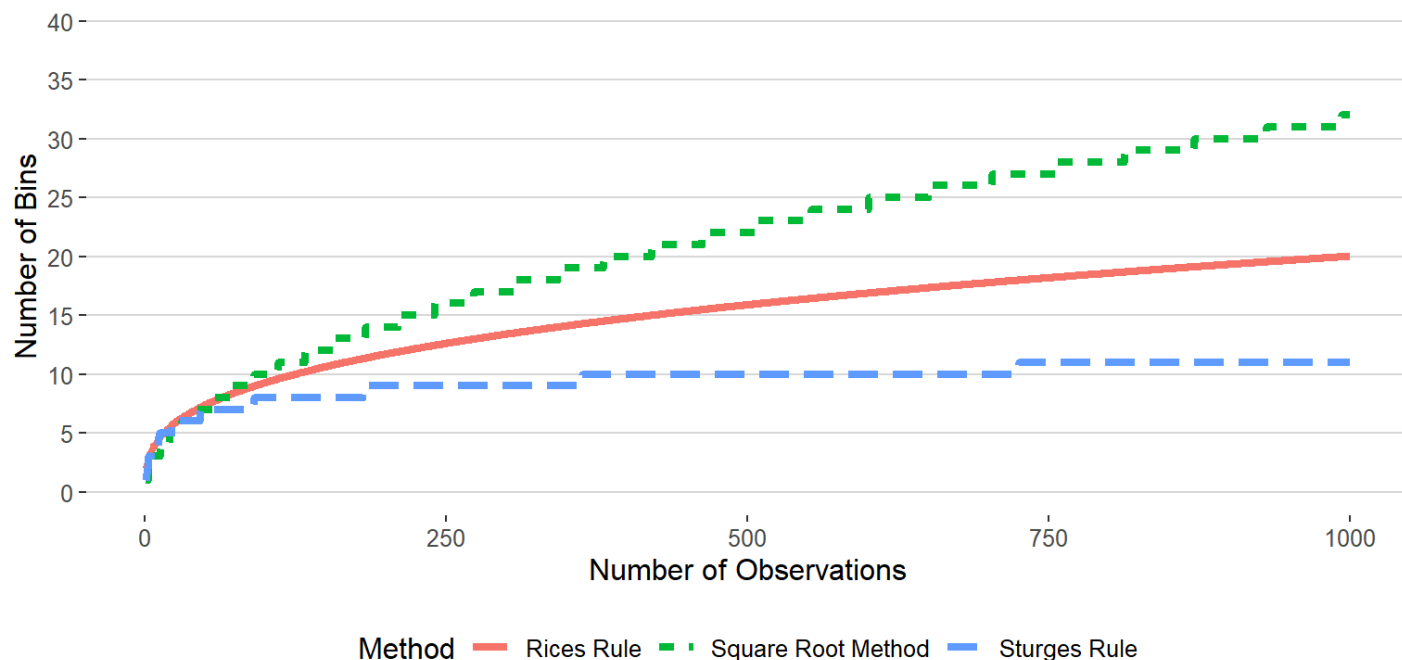
$$k = \frac{\max x - \min x}{w}$$

- formula for choosing the width based on the number of bins k

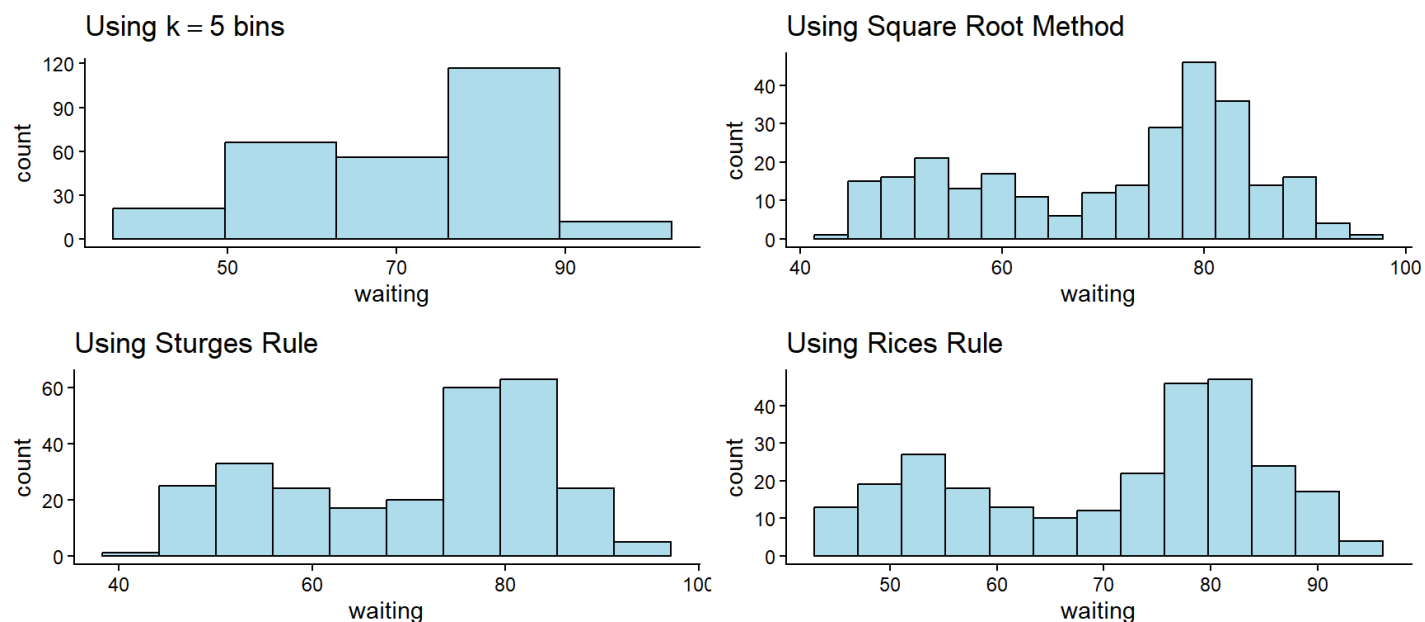
$$w = \frac{\max x - \min x}{k}$$

where $\min$ and $\max$ denote the smallest and largest values of the variable x , respectively.

- How many bins should I use? This is actually a question that has received a lot of attention in the scientific literature. Typically, you want to be sure that you don't use "too few" bins as this can cause misleading conclusions about the features of the distribution. There are several rules that are based on the coverage and number of observations of the variable:
 - **square root method** is defined as $k = \text{round}(\sqrt{n})$. This is a fairly easy and safe rule for choosing the number of bins to use. However, for large sample sizes it can lead to a very large number of bins.
 - **Sturges Rule** is defined as $k = \text{round}(\log_2 n) + 1$. This method is not great for a small number of observations (i.e $n < 30$)
 - **Rices Rule** is defined as $k = 2\sqrt[3]{n}$. This rule is a nice intermediate between Sturges Rule and the square root method.



Consider the following example of histograms constructed for the $n = 272$ observations collected on the waiting time between eruptions and the duration of the eruption of the Old Faithful geyser in Yellowstone National Park, Wyoming, USA.



Try it out: Construct a histogram for the following 10 observations of the variable X

$$X = \{-1.49, -0.65, -0.6, -0.54, -0.45, 0.01, 0.17, 0.27, 0.51, 1.34\}$$

using $k = 4$ bins/intervals

- In this example, we are given the desired number of bins, so we can use our equation to find the width of the bins w .

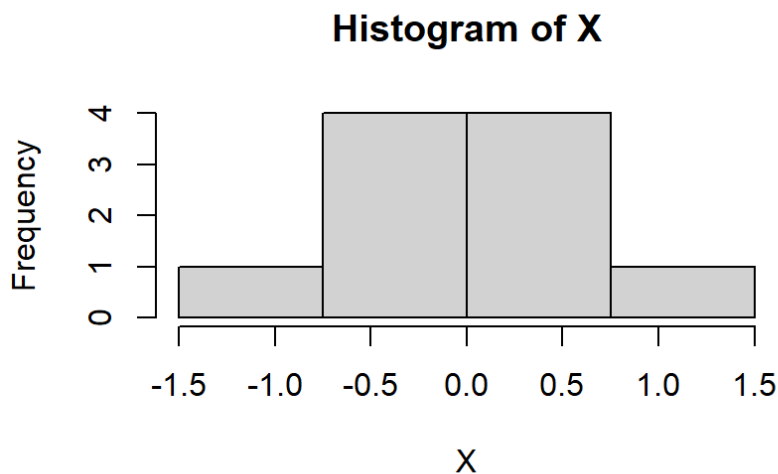
$$w = \frac{1.34 - (-1.49)}{4} = \frac{2.83}{4} = 0.7075 \approx \frac{3}{4} = 0.75$$

- now that we know about how wide each interval should be we can easily compute the intervals. First, let a be the minimum value of X and b be the maximum value such that $\min(X) = a, \max(X) = b$. The first interval is defined as $a < X \leq a + w$ which gives $-1.5 < X \leq -0.75$. The second interval is given by $-0.75 < X \leq -0.75 + w$ or $-0.75 < X \leq 0$. Continuing this pattern gives the remaining bins yields the following frequency table:

Frequency table for X using four bins of width 0.75.

X	Frequency	Relative Frequency	Cumulative Relative Frequency
$-1.5 < X \leq -0.75$	1	0.1	0.1
$-0.75 < X \leq 0$	4	0.4	0.5
$0 < X \leq 0.75$	4	0.4	0.9
$0.75 < X \leq 1.5$	1	0.1	1.0

- From the frequency table above, we can now easily construct the histogram of X by simply drawing the appropriate bars over each interval/bin:



If you'd like to read more about scaling interval widths you can check out these references:

Sturges, Herbert A. "The choice of a class interval." *Journal of the american statistical association* 21.153 (1926): 65-66. (<https://scholar.archive.org/work/l44cm5gq65f4rojkkay6cl5bq/access/wayback/http://www.esalq.usp.br/departamentos/lce/arquivos/aulas/2013/LCE0216/Sturges1926.pdf>)

Describing the shape of a distribution

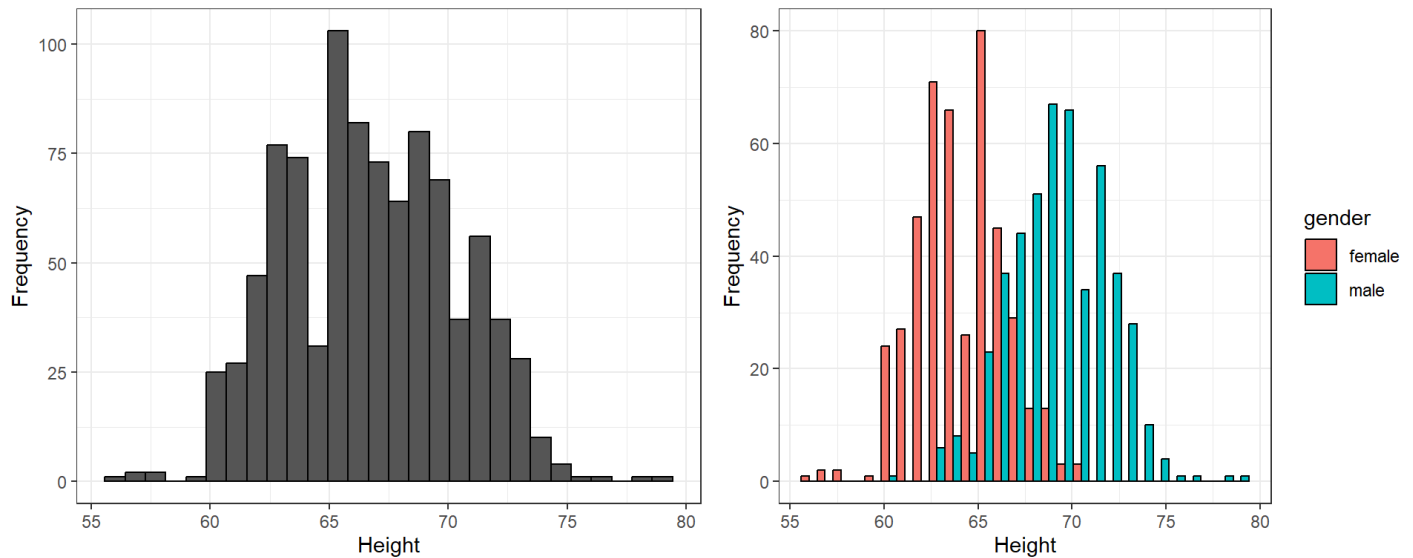
The purpose of a graph is to help us understand the data. Once the distribution of variable is plotted, you must ask yourself “what do I see?”. This is when we look for the features of a distribution such as shape, center and spread. This helps us to identify interesting patterns or potential **outliers** - individual data points that lie outside the normal pattern of the distribution.

- One of the first things we can look for are peaks of the distribution. A peak indicates values that are more likely to occur called the **mode**. A distribution can have no peaks, one peak, or multiple peaks. A distribution with one peak is said to be **unimodal**, a distribution with two peaks is **bimodal** and a distribution with multiple peaks is **multimodal**.
 - bimodal or multimodal distributions typically indicate underlying population structure in the data. Take for example the following height data from Sir Francis Galton (1822–1911) who was an English statistician. Galton was interested in the relationship between the height of parents and their offspring. He accumulated the following observations of the height of 934 children from 205 families in 1886.

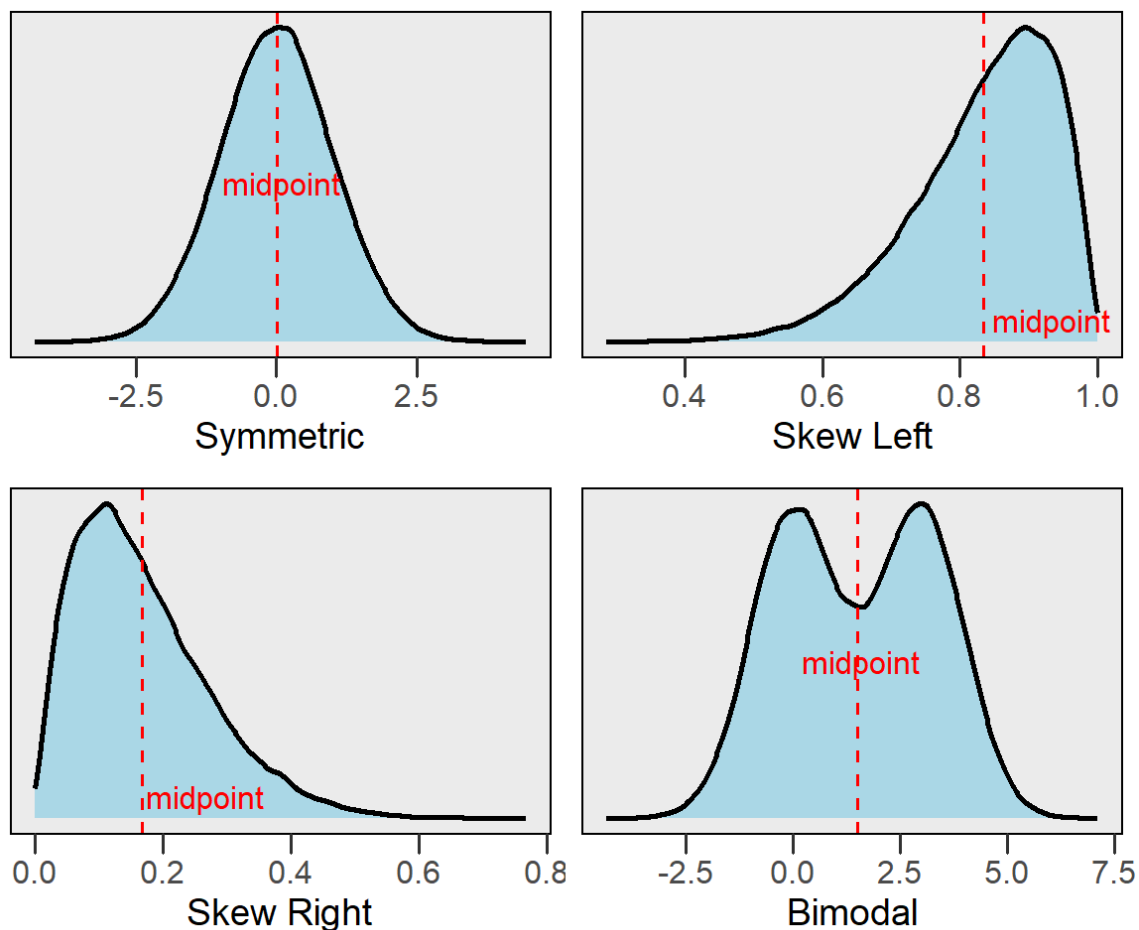
family	father	mother	midparentHeight	children	childNum	gender	childHeight
001	78.5	67	75.43	4	1	male	73.2
001	78.5	67	75.43	4	2	female	69.2
001	78.5	67	75.43	4	3	female	69
001	78.5	67	75.43	4	4	female	69
002	75.5	66.5	73.66	4	1	male	73.5
002	75.5	66.5	73.66	4	2	male	72.5
002	75.5	66.5	73.66	4	3	female	65.5
002	75.5	66.5	73.66	4	4	female	65.5
003	75	64	72.06	2	1	male	71
003	75	64	72.06	2	2	female	68

family	father	mother	midparentHeight	children	childNum	gender	childHeight
NA	:	:	:	:	:	NA	:
204	62.5	63	65.27	2	2	female	57

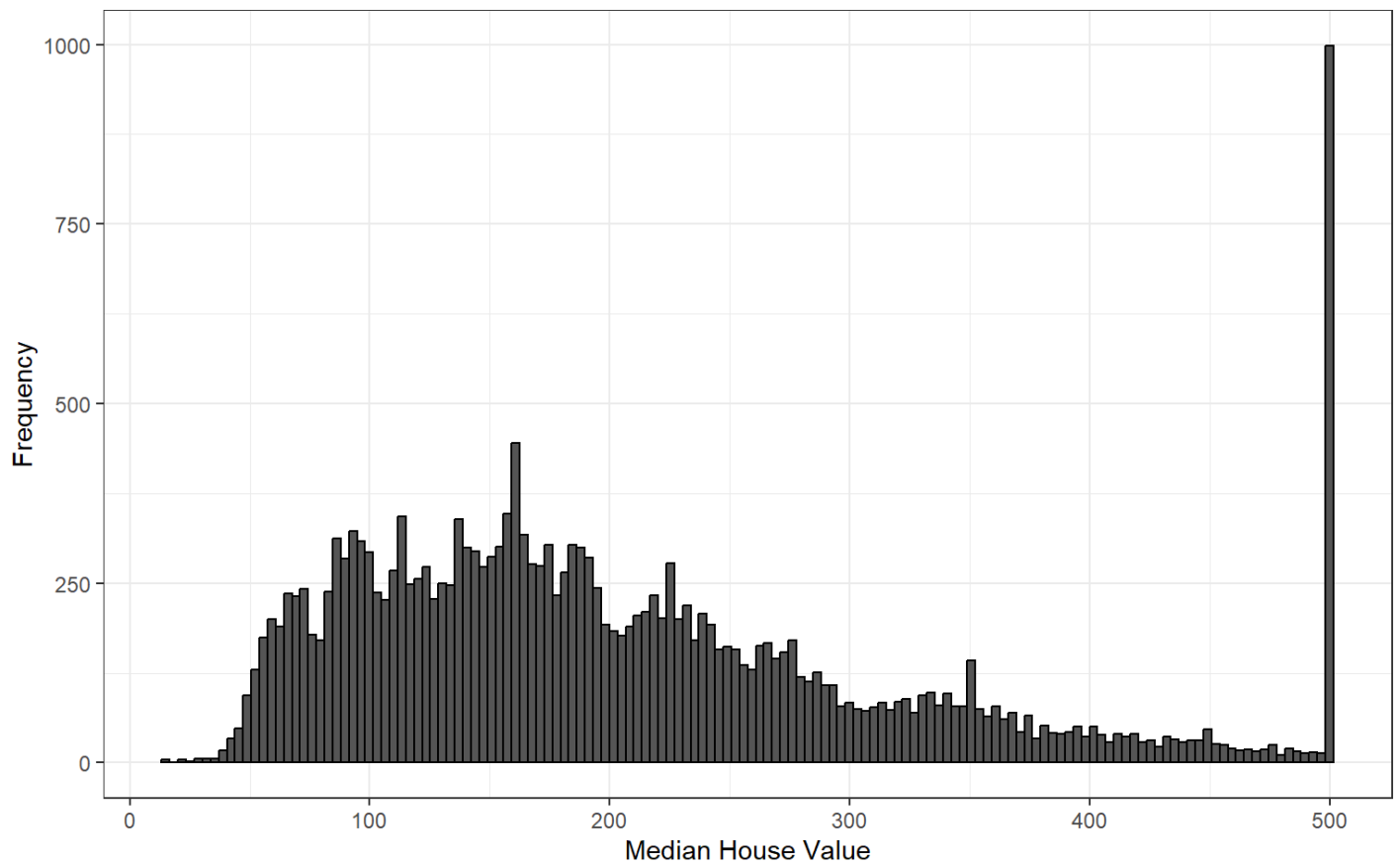
Now consider the histogram for child height:



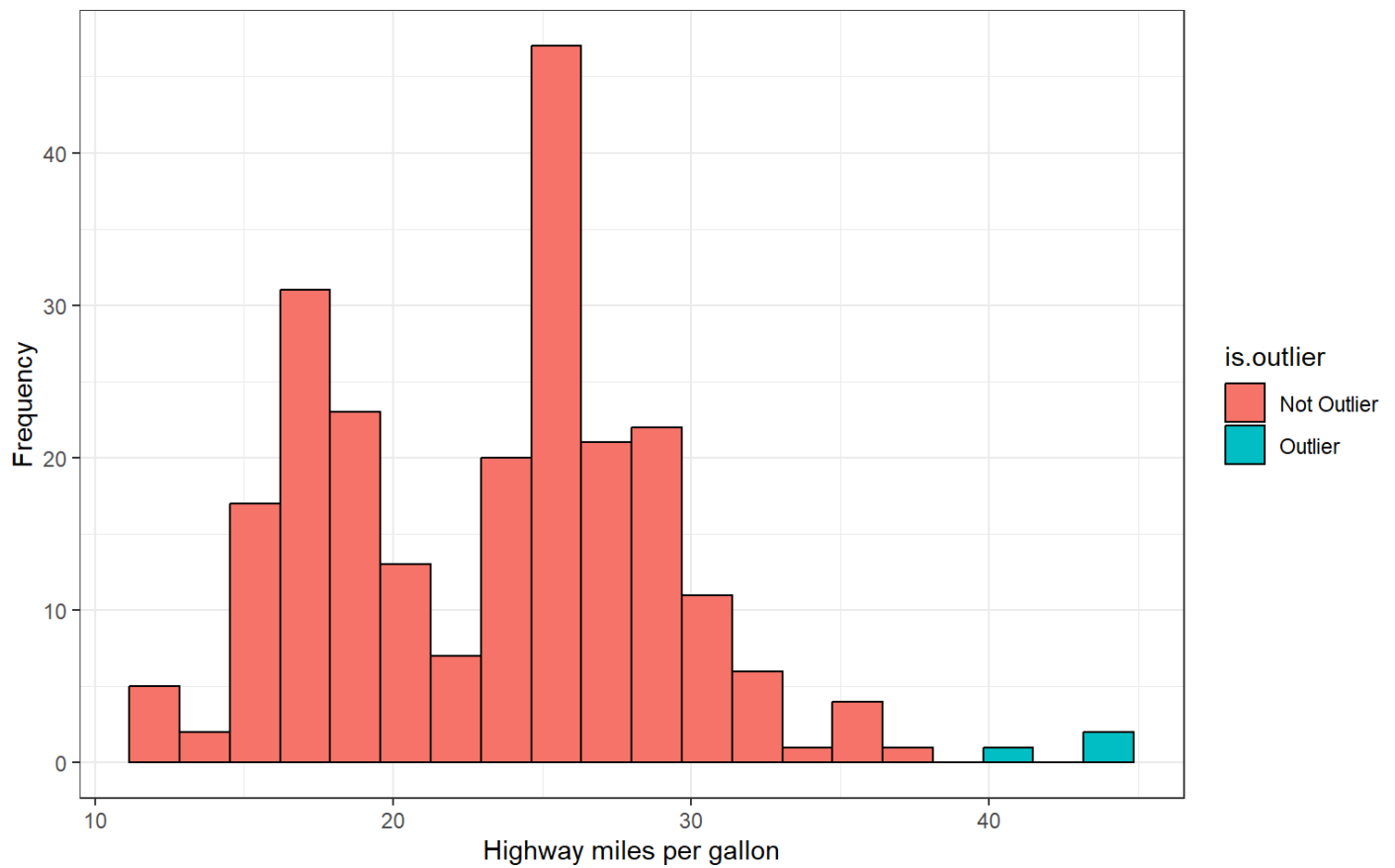
- We can see from the histogram on the left that there are two clearly visible peaks. By coloring and segregating the bars according to gender we produce the histogram on the right which shows that the two peaks correspond to the gender of the children with males being about 5 inches taller on average.
- Another question we may ask about the shape of a distribution is whether the distribution is symmetric or if it is skewed. A distribution is **symmetric** if the values that are smaller and larger than its midpoint are mirror images. A distribution is **skewed right** if the values larger than the midpoint form a long tail. Conversely, a distribution is **skewed left** if the values less than the midpoint form a long tail to the left



Many types of variables have distributions with predictable shapes. For example, variables that represent measurements of size from the same species and sex such as height, length, and age will have symmetric distributions. Variables related to money or finance on the other hand will often have skewed distributions. Skewness arises when there is a strict boundary on the values of a variable. For example, housing prices are always positive numbers and many houses will be a similar price while some houses may be really expensive. For example, consider the distribution of housing prices from the 1990 california housing census data downloaded from [Kaggle \(https://www.kaggle.com/datasets/camnugent/california-housing-prices?resource=download\)](https://www.kaggle.com/datasets/camnugent/california-housing-prices?resource=download).



Lastly, one of the most important things a graphical summary of a variable can show is **outliers**. As previously noted, these are extreme values that fall far away from the normal pattern of the data. An outlier may be a result of measurement error or it can represent a real datapoint. Either way, outliers can strongly influence any analyses we conduct later on during statistical inference and are crucial to be aware of. Consider the following histogram of fuel efficiency (measured in miles per gallon) of several different car brands between 1999 and 2008 obtained from <https://fuelconomy.gov/> (<https://fuelconomy.gov/>)



- How would you describe the shape of this distribution?
- From the histogram we can see there are two bars that fall far to the right of the rest of the distribution. The observations comprising these bars constitute potential outliers. In fact, these three observations belong to the 1999 volkswagen beetle and jetta which was extremely fuel efficient for its time